

NPDA's and CFGs have equivalent expressive power

Theorem 1. Given a CFG  $G$ , we can construct an NPDA  $M$  s.t.  $L(G) = L(M)$ .

Theorem 2. Given an NPDA  $M$ , we can construct a CFG  $G$  s.t.  $L(G) = L(M)$ .

CFG to NPDA.

Suppose  $G = (N, \Sigma, P, S)$ . To construct NPDA  $M$   
s.t.  $L(M) = L(G)$ .

Assume that all productions of  $G$  are of the form

$$A \rightarrow c B_1 B_2 \dots B_k, c \in \Sigma \cup \{\epsilon\}, k \geq 0.$$

Greibach Normal Form.

CFG to NPDA.

Suppose  $G = (N, \Sigma, P, S)$ . To construct NPDA  $M$   
s.t.  $L(M) = L(G)$ .

Assume that all productions of  $G$  are of the form

$$A \rightarrow c B_1 B_2 \dots B_k, c \in \Sigma \cup \{\epsilon\}, k \geq 0.$$

Greibach Normal Form.

Construct  $M = (\{q\}, \Sigma, N, \delta, q, S, \phi)$

$M$  has one state and accepts with empty stack

$\Sigma$  - set of terminal symbols in  $G$ : input alphabet  
in  $M$

$N$  - set of nonterminals of  $G$  is the stack alphabet of  
 $M$ .

$S$  - start symbol of  $G$ , the initial stack symbol of  
 $M$ .

Definition of  $\delta$ :

For each production  $A \rightarrow c B_1 B_2 \dots B_k$ , add  
the following to  $\delta$ .

$$((q, c, A), (q, B_1 B_2 \dots B_k))$$

$$M = (\{q\}, \Sigma, N, S, q, S, \phi)$$

$S$ : For each  $A \rightarrow C B_1 B_2 \dots B_k, ((q, C, A), (q, B_1 B_2 \dots B_k)) \in S$ .

Example. Balanced Parentheses.

Production Rules in  $G$

1.  $S \rightarrow [BS$
2.  $S \rightarrow [B$
3.  $S \rightarrow [SB$
4.  $S \rightarrow [SBS$
5.  $B \rightarrow ]$

Transitions in  $M$ .

- $((q, [, S), (q, BS))$ .
- $((q, [, S), (q, B))$
- $((q, [, S), (q, SB))$
- $((q, [, S), (q, SBS))$
- $((q, ], B), (q, \epsilon))$

Leftmost derivation - productions are always applied to the leftmost nonterminal.

$$M = (\{q\}, \Sigma, N, S, q, S, \phi)$$

$S$ : For each  $A \rightarrow C B_1 B_2 \dots B_k, ((q, C, A), (q, B_1 B_2 \dots B_k)) \in S$ .

Example. Balanced Parentheses.

Production Rules in  $G$

1.  $S \rightarrow [BS$
2.  $S \rightarrow [B$
3.  $S \rightarrow [SB$
4.  $S \rightarrow [SBS$
5.  $B \rightarrow ]$

Transitions in  $M$ .

- $((q, [, S), (q, BS))$ .
- $((q, [, S), (q, B))$
- $((q, [, S), (q, SB))$
- $((q, [, S), (q, SBS))$
- $((q, ], B), (q, \epsilon))$

Leftmost derivation - productions are always applied to the leftmost nonterminal.

To show. Leftmost derivation of  $x$  in  $G$  corresponds to an accepting computation of  $M$  on input  $x$ .

Example: Input  $x = [[[]][[]]$

Rule

Configurations of  $M$

|   |              |                             |
|---|--------------|-----------------------------|
|   | $S$          | $(q, [[[]][[]], S)$         |
| 3 | $[SB$        | $(q, [[]][[]], SB)$         |
| 4 | $[[SBSB$     | $(q, []][[]], SBSB)$        |
| 2 | $[[[BB]SB$   | $(q, ]][[]], BB]SB)$        |
| 5 | $[[[[]]BSB$  | $(q, ]][[]], BSB)$          |
| 5 | $[[[[]]]SB$  | $(q, [[]], SB)$             |
| 2 | $[[[[]]]BB$  | $(q, []], BB)$              |
| 5 | $[[[[]]][]B$ | $(q, ], B)$                 |
| 5 | $[[[[]]][]]$ | $(q, \epsilon, \epsilon)$ . |

Lemma 1 . For any  $z, y \in \Sigma^*$ ,  $\gamma \in N^*$  and  $A \in N$ ,

$A \xrightarrow{n} z\gamma$  by a leftmost derivation iff  $(q, zy, A) \xrightarrow{n} (q, y, \gamma)$

Lemma 1. For any  $z, y \in \Sigma^*$ ,  $\gamma \in N^*$  and  $A \in N$ ,

$A \xrightarrow{n}_G z\gamma$  by a leftmost derivation iff  $(q, zy, A) \xrightarrow{n}_M (q, y, \gamma)$

Proof. Induction on  $n$ .

Base case,  $n=0$ : Straight forward.

Induction Step.

Suppose  $A \xrightarrow{n+1}_G z\gamma$  using a leftmost derivation.

$\rightarrow C \in \Sigma \cup \{\epsilon\}, \beta \in N^*$

Suppose  $B \rightarrow C\beta$  was the last production applied.

$A \xrightarrow{n}_G uB\alpha \xrightarrow{1}_G uC\beta\alpha = z\gamma$ . /  $z = uC$  and  $\gamma = \beta\alpha$

By induction hypothesis,  $(q, ucy, A) \xrightarrow{n}_M (q, cy, B\alpha)$ .

By definition of  $M$ ,  $((q, c, B), (q, \beta)) \in \delta$ .

therefore,  $(q, cy, B\alpha) \xrightarrow{1}_M (q, y, \beta\alpha)$

Thus we have

$(q, zy, A) = (q, ucy, A) \xrightarrow{n+1}_M (q, y, \beta\alpha) = (q, y, \gamma)$ .

Conversely, Suppose  $(q, zy, A) \xrightarrow{n+1}_M (q, y, \gamma)$

let  $((q, cy, B), (q, \beta)) \in S$  be the last transition taken by  $M$ .

Then  $z = uc$  for some  $u \in \Sigma^*$ ,  $\gamma = \beta\alpha$  for some  $\alpha \in \Gamma^*$  and

$$(q, ucy, A) \xrightarrow{n}_M (q, cy, B\alpha) \xrightarrow{1}_M (q, y, \beta\alpha)$$

Conversely, Suppose  $(q, zy, A) \xrightarrow{n+1}_M (q, y, \gamma)$

let  $((q, cB), (q, \beta)) \in S$  be the last transition taken by  $M$ .

Then  $z = uc$  for some  $u \in \Sigma^*$ ,  $\gamma = \beta\alpha$  for some  $\alpha \in \Gamma^*$  and

$$(q, ucy, A) \xrightarrow{n}_M (q, cy, B\alpha) \xrightarrow{1}_M (q, y, \beta\alpha)$$

By induction hypothesis  $A \xrightarrow{n}_G uB\alpha$  by a leftmost derivation in  $G$ .

By definition of  $S$  in  $M$ ,  $B \rightarrow cB$  is a production of  $G$ .

Then,  $A \xrightarrow{n}_G uB\alpha \xrightarrow{1}_G uc\beta\alpha = z\gamma$  by a leftmost derivation.

Conversely, Suppose  $(q, zy, A) \xrightarrow{M^{n+1}} (q, y, \gamma)$

let  $((q, c, B), (q, \beta)) \in S$  be the last transition taken by  $M$ .

Then  $z = uc$  for some  $u \in \Sigma^*$ ,  $\gamma = \beta\alpha$  for some  $\alpha \in \Gamma^*$  and

$$(q, ucy, A) \xrightarrow{M^n} (q, cy, B\alpha) \xrightarrow{M^1} (q, y, \beta\alpha)$$

By induction hypothesis  $A \xrightarrow{G^n} uB\alpha$  by a leftmost derivation in  $G$ .

By definition of  $S$  in  $M$ ,  $B \rightarrow c\beta$  is a production of  $G$ .

Then,  $A \xrightarrow{G^n} uB\alpha \xrightarrow{G^1} uc\beta\alpha = z\gamma$  by a leftmost derivation.

Theorem  $L(G) = L(M)$ .

Proof.

$x \in L(G)$  iff  $S \xrightarrow{G^*} x$  by a leftmost derivation  
[Defn of  $L(G)$ ]

iff  $(q, x, S) \xrightarrow{M^*} (q, \epsilon, \epsilon)$  [Lemma 1]

iff  $x \in L(M)$  [Definition of  $L(M)$ ]

NPDA's accept only CFLs.

Step 1. Every PDA can be simulated by a PDA with one state.

Step 2. Every PDA with one state has an equivalent CFG.

NPDA's accept only CFL's.

Step 1. Every PDA can be simulated by a PDA with one state.

Step 2. Every PDA with one state has an equivalent CFG.

→ "Invert" the construction in the previous lecture.

Suppose  $M = (\{q\}, \Sigma, \Gamma, S, \delta, \perp, \phi)$

Define  $G = (\Gamma, \Sigma, P, \perp)$  as follows.

For every transition  $((q, c, A), (q, B_1 \dots B_k)) \in \delta$ ,

add the production  $A \rightarrow cB_1B_2 \dots B_k$  in  $P$ .

Lemma 1. For any  $z, y \in \Sigma^*$ ,  $x \in N^*$  and  $A \in N$ ,

$A \xrightarrow{n}_G zyx$  by a leftmost derivation iff  $(q, zyx, A) \xrightarrow{n}_M (q, y, x)$

Theorem  $L(G) = L(M)$ .